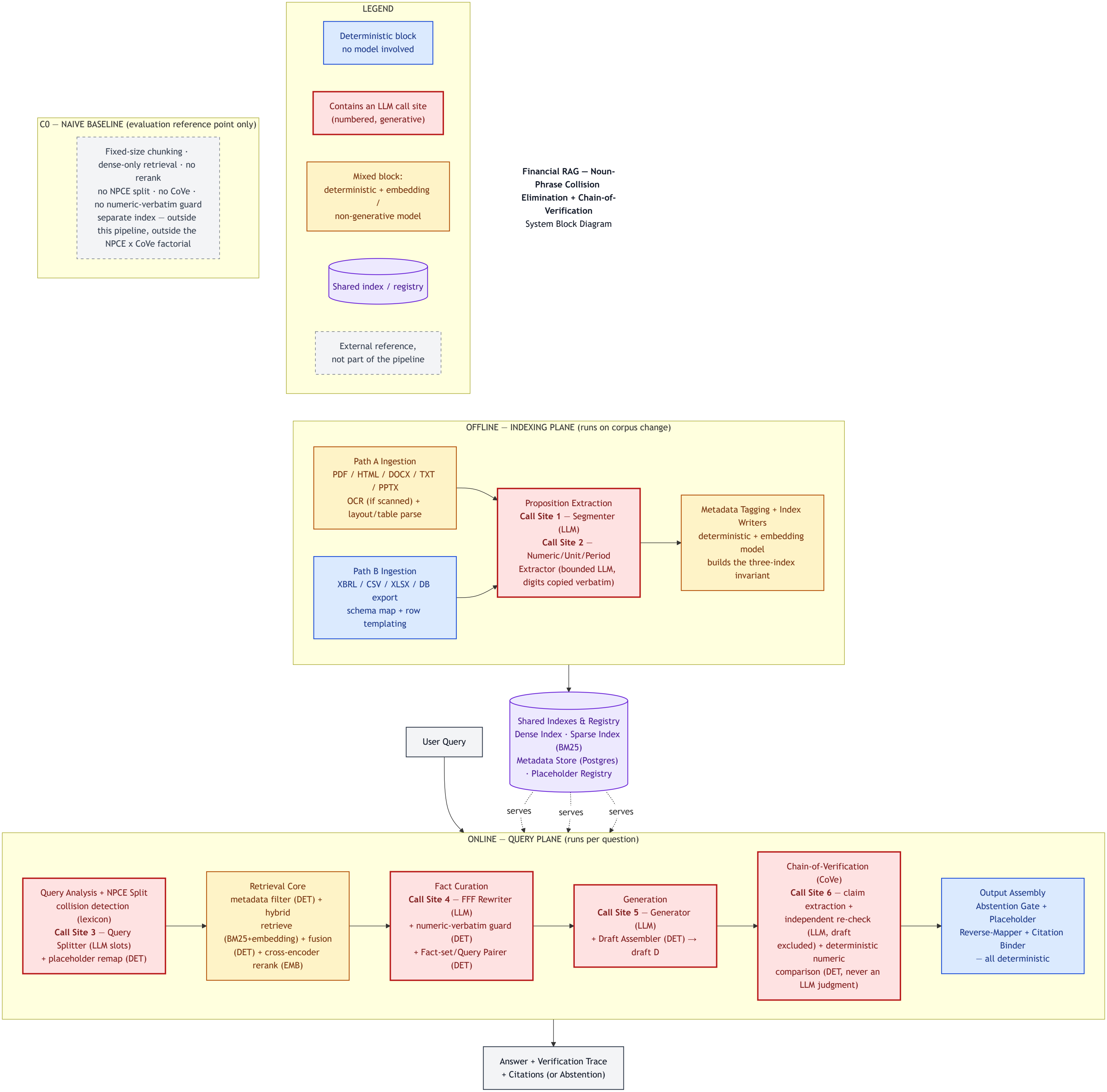
Financial RAG — Noun-Phrase Collision Elimination + Chain-of-Verification
Full System Architecture: Offline Indexing Plane + Online Query Plane

LEGEND
Deterministic code no model involved
LLM call — generative numbered = one of the six permitted call sites
Embedding / cross-encoder — non-generative neural model
Mixed: deterministic code + bounded/partial model use
Index / data store
Architectural note (on/off behaviour)
C0 — NAIVE BASELINE (evaluation reference point only)
Fixed-size chunking (2000 chars, per page) · dense-only retrieval · no rerank · no Query Splitter · no Metadata Filter · no CoVe · no numeric-verbatim guard · separate index (financial_rag_c0)
Outside the NPCE x CoVe factorial — not part of the pipeline above.
OFFLINE PLANE — runs on corpus change
Path A inputs: PDF / HTML / DOCX / TXT / PPTX
Scanned — no text layer?
yes
no
OCR + table structure recovery PaddleOCR PP-Structure
Layout- & Table-Aware Parser format-driven, content-agnostic
Proposition Segmenter **CALL SITE 1 — LLM** DenseX-style atomic segmentation structured output, may not alter digits
Path B inputs: XBRL / CSV / XLSX / DB export
Schema Mapper maps columns/tags to canonical fields
Row-to-Proposition Templating string templating, no LLM
Numeric / Unit / Currency / Period Extractor **CALL SITE 2 — bounded LLM** tier1/2 regex first; LLM resolves qualifiers only — digits copied verbatim by code
Metadata Tagger from filing headers / XBRL tags
Three-index invariant: Dense, Sparse, Metadata Store index identical proposition_id sets
Index Writers deterministic + embedding model
Placeholder Registry Builder deterministic + embedding similarity clustering
Dense Index
Sparse Index — BM25
Metadata Store — Postgres/JSONB
Placeholder Registry
serves
serves
serves
serves
serves
ONLINE PLANE — runs per question
User query
Query Analyzer noun-phrase extraction + collision detection (lexicon-only)
Query Splitter **CALL SITE 3 — LLM (slot parsing)** + DET (Cartesian product = the split itself)
Query-side Placeholder Remapper dictionary substitution
NPCE off (C1 / C2): exactly one unsplit sub-query — passthrough, not a code branch
Fan out: one branch per sub-query s_i
Metadata Filter SQL over Postgres
Hybrid Retriever BM25 (det) + Dense (embedding)
Rank Fusion + time-aware weighting
Cross-encoder Reranker non-generative model
FFF Rewriter **CALL SITE 4 — LLM** + DET numeric-verbatim guard (rejects any unsourced digit)
Fact-set / Query Pairer keyed binding only
Generator **CALL SITE 5 — LLM** answers from paired FFFs only
CoVe off (C1 / C3): draft D passes straight to the Abstention Gate
Draft Assembler templating; introduces no new facts → draft D
Chain-of-Verification — CALL SITE 6
Extract atomic claims from draft D **6a — LLM**
Template verification questions DET — deterministic construction
Execute verification independently **6b — LLM** draft D structurally absent from this call's context — not a prompt instruction
Claim type?
numeric
prose
Deterministic numeric comparison DET — value equality **never an LLM judgment**
Prose comparison **6c — LLM**
Revised / flagged claim set
Abstention Gate rules over verification results
Placeholder Reverse-Mapper restores real entity names
Citation Binder looks up proposition_id metadata **an LLM never writes a citation**
Answer + verification trace + citations (or abstention)



Financial RAG — Noun-Phrase Collision Elimination + Chain-of-Verification
System Block Diagram

LEGEND
Deterministic block
no model involved
Contains an LLM call site
(numbered, generative)
Mixed block:
deterministic + embedding / non-generative model
Shared index / registry
External reference,
not part of the pipeline
C0 — NAIVE BASELINE (evaluation reference point only)
Fixed-size chunking · dense-only retrieval · no rerank
no NPCE split · no CoVe · no numeric-verbatim guard
separate index — outside this pipeline, outside the NPCE x CoVe factorial
OFFLINE — INDEXING PLANE (runs on corpus change)
Path A Ingestion
PDF / HTML / DOCX / TXT / PPTX
OCR (if scanned) + layout/table parse
Path B Ingestion
XBRL / CSV / XLSX / DB export
schema map + row templating
Proposition Extraction
Call Site 1 — Segmenter (LLM)
Call Site 2 — Numeric/Unit/Period Extractor (bounded LLM, digits copied verbatim)
Metadata Tagging + Index Writers
deterministic + embedding model
builds the three-index invariant
Shared Indexes & Registry
Dense Index · Sparse Index (BM25)
Metadata Store (Postgres) · Placeholder Registry
User Query
serves
serves
serves
ONLINE — QUERY PLANE (runs per question)
Query Analysis + NPCE Split
collision detection (lexicon)
Call Site 3 — Query Splitter (LLM slots)
+ placeholder remap (DET)
Retrieval Core
metadata filter (DET) + hybrid
retrieve (BM25+embedding) + fusion (DET) + cross-encoder rerank (EMB)
Fact Curation
Call Site 4 — FFF Rewriter (LLM)
+ numeric-verbatim guard (DET)
+ Fact-set/Query Pairer (DET)
Generation
Call Site 5 — Generator (LLM)
+ Draft Assembler (DET) → draft D
Chain-of-Verification (CoVe)
Call Site 6 — claim extraction + independent re-check (LLM, draft excluded) + deterministic numeric comparison (DET, never an LLM judgment)
Output Assembly
Abstention Gate + Placeholder Reverse-Mapper + Citation Binder
— all deterministic
Answer + Verification Trace
+ Citations (or Abstention)

Financial RAG — Entity Relationship Diagram (contracts/models.py frozen contracts + evaluation gold key)

resolved into placeholder_map
verification_trace
verification_trace
fff_sets, keyed by sub_query_id
collision_tags
sub_queries
evidence points into indexed pages
gold_answers
expected_sub_queries (ground truth split)
source.evidence_refs
supporting_fff_ids
rewritten into (Call Site 4)
numeric (optional)
scored against Query Splitter output
source_ref
period
period

PLACEHOLDER_REGISTRY_ENTRY

string	placeholder	PK	e.g. ENT_1
string	surface_form		
string	ticker		
string	cik		

REQUEST_CONTEXT

string	request_id	PK	
string	config_hash		
string	placeholder_map		dict: placeholder to surface_form, per-request only, never global

VERIFIED_ANSWER

string	answer_text

COLLISION_RECORD

string	collision_id	PK	
string	dataset		FinanceBench-Collision primary or secondary
string	collision_axis		temporal | entity | metric
string	question		
string	unit		
string	currency		
string	scale		
string	verification_status		
bool	requires_calculation		
bool	convention_ambiguous		
string	construction_rule		R1_temporal | R2_metric | R3_entity

VERIFICATION_CLAIM_RESULT

string	claim_id	PK	
string	claim_text		
bool	is_numeric		
string	verdict		match | conflict | unsupported
string	supporting_fff_ids		tuple of FFF ids

COLLISION_TAG

string	axis	temporal | entity | metric
string	members	tuple of colliding items

PROPOSITION

string	proposition_id	PK	
string	text		
string	provenance_tier		retrieved | derived | external
string	ingestion_path		A_unstructured | B_structured
float	ocr_confidence		
string	extraction_tier		xbrl | rules | llm_qualifier
string	company		
string	ticker		
string	filing_type		
string	filing_date		
string	section		
int	page		zero-indexed, §8A.1
string	source_uri		

GOLD_ANSWER

string	period_key
float	value

EXPECTED_SUB_QUERY

string	company
string	metric
string	period

EVIDENCE_REF

string	doc_name	nullable, resolves to primary_doc
int	page_num_zero_indexed	

FFF

string	fff_id	PK	
string	proposition_id	FK	
string	subject		placeholder token, e.g. ENT_1
string	metric		
string	period_label		
string	value_text		
string	provenance_tier		

SUB_QUERY

string	sub_query_id	PK	
string	entity		
string	metric		
string	period_label		
string	filing_types		tuple, optional filter

NUMERIC_RECORD

float	value	always copied verbatim from source, never LLM-generated
string	scale	unit | thousands | millions | billions
string	currency	
string	unit	percent | ratio | absolute | count

SOURCE_REF

string	filing_type	
string	filing_date	
int	page	joined deterministically, never model output

PERIOD

int	fiscal_year	
int	quarter	1-4, optional
string	basis	reported | ttm | restated

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Flowchart: One Query's Path Through the Online Query Plane

LEGEND
Deterministic step no model involved
LLM call — generative numbered = one of the six permitted call sites
Embedding / cross-encoder — non-generative model
Mixed: deterministic + embedding model
Decision point
User submits a question
Query Analyzer noun-phrase extraction + collision detection (lexicon-only)
NPCE enabled for this run?
off — C1 / C2
on — C3 / C4
Query Splitter **CALL SITE 3 — LLM** (slot parsing only) + deterministic Cartesian product (the split itself is code, not the model)
Single pass-through sub-query — no split, identity transform
Query-side Placeholder Remap each resulting sub-query names exactly one entity, one metric, one period
Fan out: N sub-queries (N = 1 if NPCE off)
Per sub-query s_i — repeated independently for each of the N sub-queries
Metadata Filter SQL over Postgres — narrows candidates by this sub-query's entity/period/metric tags
Hybrid Retrieve BM25 (det) + Dense (embedding)
Rank Fusion + time-aware weighting
Cross-encoder Rerank non-generative model
FFF Rewriter **CALL SITE 4 — LLM** rewrites top propositions into Fully-Formatted Facts
Numeric-verbatim guard: does every digit & period trace verbatim to the source proposition?
passes
fails
Fact-set / Query Pairer keyed binding only
Generator **CALL SITE 5 — LLM** answers from this sub-query's paired FFFs only
Reject this FFF — metric honestly reported 'not provided'
Sub-answer a_i
Draft Assembler DET — collects all N sub-answers, introduces no new facts → draft D
CoVe enabled for this run?
off — C1 / C3
on — C2 / C4
Extract atomic claims from draft D **CALL SITE 6a — LLM**
Template one verification question per claim DET — deterministic construction
Execute verification independently **CALL SITE 6b — LLM** draft D is structurally absent from this call's context — not a prompt instruction
Claim type?
numeric
prose
Deterministic numeric comparison DET — value equality, never an LLM judgment
Prose comparison **CALL SITE 6c — LLM**
Draft D passes straight through, unverified
Per-claim verdict: match / conflict / unsupported
Abstention Gate DET — rules over verification results (all-or-nothing across a question's claims)
Every required claim verified & evidence-supported?
no
yes
Abstain — 'not found in source', no citation, no guess
Placeholder Reverse-Mapper DET — restores real entity names
Citation Binder DET — looks up proposition_id metadata **an LLM never writes a citation**
Final answer + citations + verification trace